

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: Use FTP Services

Prepared by:
Kudzaishe Majeza
Junior Network Engineer – VxLEARN Networks

Mentor:
Titus Majeza
Senior Network Engineer

Date: 19 November 2025

Table of Contents

1. Introduction

2. Objectives

3. Addressing Table

4. Background / Scenario

5. Part 1 – Upload a File to an FTP Server

5.1 Step 1: Locate the File

5.2 Step 2: Connect to the FTP Server

5.3 Step 3: Upload the File

6. Part 2 – Download a File from an FTP Server

6.1 Step 1: Rename the File

6.2 Step 2: Download the File

6.3 Step 3: Delete the File

7. Understanding FTP Concepts

8. Conclusion

1. Introduction

This lab demonstrates how to use the File Transfer Protocol (FTP) to upload, download, rename, and delete files on a remote server. FTP is widely used for transferring files between systems, especially in environments where configuration files, web content, or backups need to be moved securely and efficiently. In this exercise, I worked from PC-A and interacted with an FTP server using command-line instructions, giving me hands-on experience with real FTP operations and how control and data connections are used in the process.

2. Objectives

- Upload a file to an FTP server
- Download a file from an FTP server
- Rename a file on the FTP server
- Delete a file from the FTP server
- Practice common FTP client commands such as **put**, **get**, **dir**, **rename**, and **delete**

3. Addressing Table

Device	Interface	IP Address	Subnet Mask
FTP Server	NIC	209.165.200.226	255.255.255.224

4. Background / Scenario

FTP operates using two communication channels:

- **Port 21:** Command control channel
- **Port 20:** Data transfer channel

The client connects to the FTP server through port 21, authenticates, and issues commands such as changing directories, listing files, uploading, and downloading. When a file needs to be transferred, the server opens a separate data connection on port 20.

In this activity, I located a sample text file on the PC, uploaded it to the server, renamed it, downloaded it back to the PC, and finally deleted it from the server.



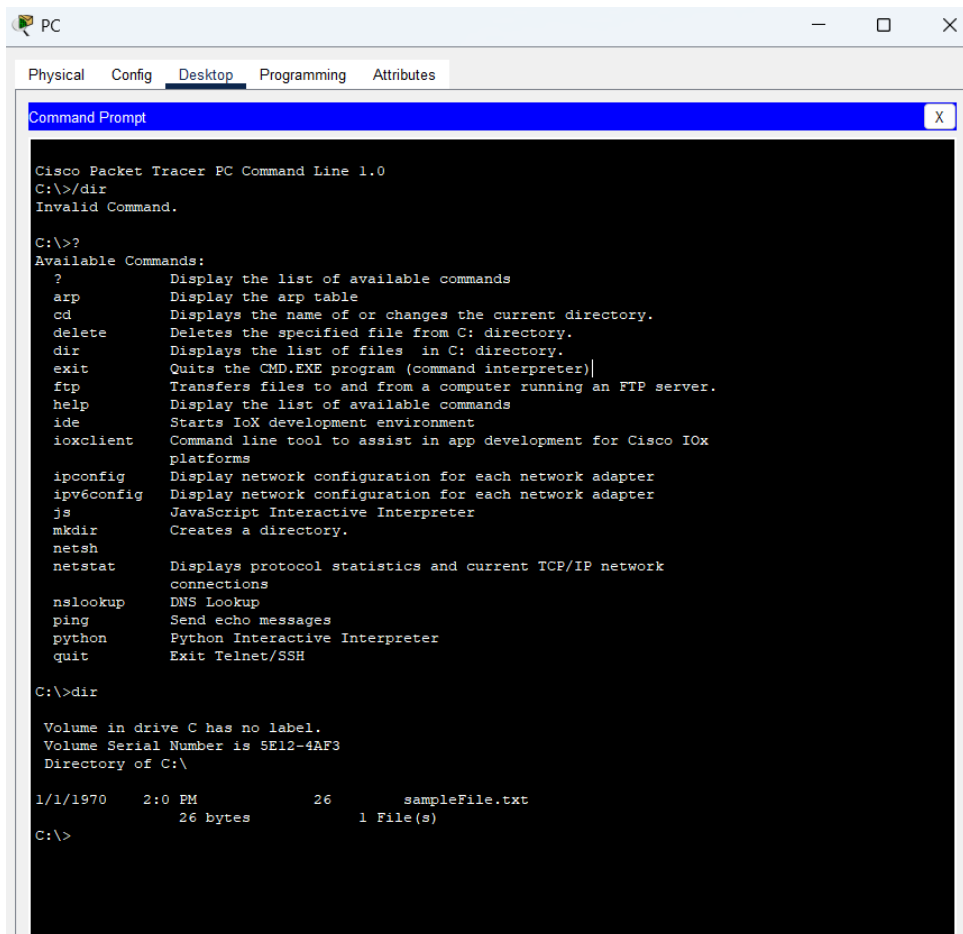
5. Part 1 – Upload a File to an FTP Server

5.1 Step 1: Locate the File

1. Open **PC-A** → **Desktop** → **Command Prompt**
2. Enter:

`dir`

3. I confirmed the file **sampleFile.txt** is stored in the C:\ directory.



```
PC
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>dir
Invalid Command.

C:\>?
Available Commands:
?          Display the list of available commands
arp        Display the arp table
cd         Displays the name of or changes the current directory.
delete     Deletes the specified file from C: directory.
dir        Displays the list of files in C: directory.
exit       Quits the CMD.EXE program (command interpreter)
ftp        Transfers files to and from a computer running an FTP server.
help       Display the list of available commands
ide        Starts IoX development environment
ioxcli     Command line tool to assist in app development for Cisco IOx
           platforms
ipconfig   Display network configuration for each network adapter
ipv6config Display network configuration for each network adapter
js         JavaScript Interactive Interpreter
mkdir     Creates a directory.
netsh      Displays protocol statistics and current TCP/IP network
           connections
nslookup   DNS Lookup
ping       Send echo messages
python     Python Interactive Interpreter
quit       Exit Telnet/SSH

C:\>dir

Volume in drive C has no label.
Volume Serial Number is 5E12-4AF3
Directory of C:\

1/1/1970    2:0 PM                26      sampleFile.txt
                26 bytes                1 File(s)

C:\>
```

5.2 Step 2: Connect to the FTP Server

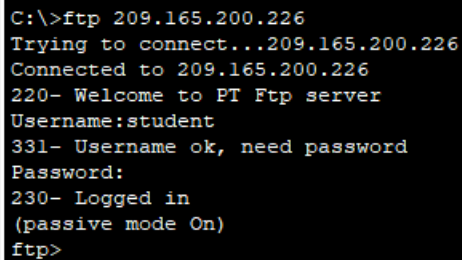
I connected to the FTP server using:

`ftp 209.165.200.226`

Login credentials:

- **Username:** student
- **Password:** class

I received a “Logged in” message confirming access.



```
C:\>ftp 209.165.200.226
Trying to connect...209.165.200.226
Connected to 209.165.200.226
220- Welcome to PT Ftp server
Username:student
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

When typing a password in the Command Prompt (or in FTP, SSH, Linux terminals, etc.):

The characters do NOT appear on screen. Not even dots ••• or stars **** show up.

It looks like nothing is being typed.

This is a security feature so nobody can see how long your password is.

So just type the password normally and press ENTER.

Even though you don't see anything, the terminal is receiving your keystrokes.

5.3 Step 2: Connect to the FTP Server

I checked the available FTP commands:

?

```
ftp>?  
?  
cd  
delete  
dir  
get  
help  
passive  
put  
pwd  
quit  
rename
```

I listed server files:

dir

```
ftp>dir  
Listing /ftp directory from 209.165.200.226:  
0 : asa842-k8.bin 5571584  
1 : asa923-k8.bin 30468096  
2 : cl841-advipservicesk9-mz.124-15.T1.bin 33591768  
3 : cl841-ipbase-mz.123-14.T7.bin 13832032  
4 : cl841-ipbasek9-mz.124-12.bin 16599160  
5 : cl900-universalk9-mz.SPA.155-3.M4a.bin 33591768  
6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768  
7 : c2600-i-mz.122-28.bin 5571584  
8 : c2600-ipbasek9-mz.124-8.bin 13169700  
9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004  
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768  
11 : c2800nm-ipbase-mz.123-14.T7.bin 5571584  
12 : c2800nm-ipbasek9-mz.124-8.bin 15522644  
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768  
14 : c2950-i6q412-mz.121-22.EA4.bin 3058048  
15 : c2950-i6q412-mz.121-22.EA8.bin 3117390  
16 : c2960-lanbase-mz.122-25.FX.bin 4414921  
17 : c2960-lanbase-mz.122-25.SEE1.bin 4670455  
18 : c2960-lanbasek9-mz.150-2.SE4.bin 4670455  
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192  
20 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279  
21 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768  
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236  
23 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849  
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552  
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138  
26 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869  
27 : ir800-universalk9-mz.SPA.155-3.M 61750062  
28 : ir800-universalk9-mz.SPA.156-3.M 63753767  
29 : ir800_yocto-1.7.2.tar 2877440  
30 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000  
31 : pt1000-i-mz.122-28.bin 5571584  
32 : pt3000-i6q412-mz.121-22.EA4.bin 3117390  
ftp>
```

Then uploaded the file:

```
put sampleFile.txt
```

```
ftp>put sampleFile.txt

Writing file sampleFile.txt to 209.165.200.226:
File transfer in progress...

[Transfer complete - 26 bytes]

26 bytes copied in 0.092 secs (282 bytes/sec)
ftp>
```

The transfer finished successfully.

I confirmed it by running `dir` again on the server.

```
ftp>dir

Listing /ftp directory from 209.165.200.226:
 0 : asa842-k8.bin                    5571584
 1 : asa923-k8.bin                    30468096
 2 : cl841-advipservicesk9-mz.124-15.T1.bin 33591768
 3 : cl841-ipbase-mz.123-14.T7.bin      13832032
 4 : cl841-ipbasek9-mz.124-12.bin      16599160
 5 : cl900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 7 : c2600-i-mz.122-28.bin             5571584
 8 : c2600-ipbasek9-mz.124-8.bin        13169700
 9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin    5571584
12 : c2800nm-ipbasek9-mz.124-8.bin      15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2950-i6q412-mz.121-22.EA4.bin    3058048
15 : c2950-i6q412-mz.121-22.EA8.bin    3117390
16 : c2960-lanbase-mz.122-25.FX.bin     4414921
17 : c2960-lanbase-mz.122-25.SEE1.bin   4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin    4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
27 : ir800-universalk9-mz.SPA.155-3.M    61750062
28 : ir800-universalk9-mz.SPA.156-3.M    63753767
29 : ir800_yocto-1.7.2.tar              2877440
30 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
31 : pt1000-i-mz.122-28.bin            5571584
32 : pt3000-i6q412-mz.121-22.EA4.bin    3117390
33 : sampleFile.txt                    26
ftp>
```


6. Part 2 – Download a File from an FTP Server

6.1 Step 1: Rename the File on the Server

At the FTP prompt, I renamed the file:

```
rename sampleFile.txt sampleFile_FTP.txt
```

```
ftp>rename sampleFile.txt sampleFile_FTP.txt
Renaming sampleFile.txt
ftp>
[OK Renamed file successfully from sampleFile.txt to sampleFile_FTP.txt]
ftp>
```

I verified the rename using:

```
dir
```

```
ftp>dir
Listing /ftp directory from 209.165.200.226:
0   : asa842-k8.bin                5571584
1   : asa923-k8.bin                30468096
2   : cl841-advipservicesk9-mz.124-15.T1.bin  33591768
3   : cl841-ipbase-mz.123-14.T7.bin  13832032
4   : cl841-ipbasek9-mz.124-12.bin  16599160
5   : cl900-universalk9-mz.SPA.155-3.M4a.bin  33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin  33591768
7   : c2600-i-mz.122-28.bin        5571584
8   : c2600-ipbasek9-mz.124-8.bin   13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin  33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin  5571584
12  : c2800nm-ipbasek9-mz.124-8.bin  15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin  33591768
14  : c2950-i6q4l2-mz.121-22.EA4.bin  3058048
15  : c2950-i6q4l2-mz.121-22.EA8.bin  3117390
16  : c2960-lanbase-mz.122-25.FX.bin  4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin  4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin  4670455
19  : c3560-advipservicesk9-mz.122-37.SEE1.bin  8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin  10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin  33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin  83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin  505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG  159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG  184530138
26  : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869
27  : ir800-universalk9-mz.SPA.155-3.M  61750062
28  : ir800-universalk9-mz.SPA.156-3.M  63753767
29  : ir800_yocto-1.7.2.tar        2877440
30  : ir800_yocto-1.7.2_python-2.7.3.tar  6912000
31  : pt1000-i-mz.122-28.bin        5571584
32  : pt3000-i6q4l2-mz.121-22.EA4.bin  3117390
33  : sampleFile_FTP.txt          26
ftp>
```

6.2 Step 2: Download the File

To download the renamed file, I used:

```
get sampleFile_FTP.txt
```

```
ftp>get sampleFile_FTP.txt

Reading file sampleFile_FTP.txt from 209.165.200.226:
File transfer in progress...

[Transfer complete - 26 bytes]

26 bytes copied in 0.01 secs (2600 bytes/sec)
ftp>
```

The file was saved locally.

I confirmed the download with another:

```
ftp>dir

Listing /ftp directory from 209.165.200.226:
0  : asa842-k8.bin                               5571584
1  : asa923-k8.bin                               30468096
2  : c1841-advipservicesk9-mz.124-15.T1.bin      33591768
3  : c1841-ipbase-mz.123-14.T7.bin               13832032
4  : c1841-ipbasek9-mz.124-12.bin                16599160
5  : c1900-universalk9-mz.SPA.155-3.M4a.bin      33591768
6  : c2600-advipservicesk9-mz.124-15.T1.bin      33591768
7  : c2600-i-mz.122-28.bin                       5571584
8  : c2600-ipbasek9-mz.124-8.bin                 13169700
9  : c2800nm-advipservicesk9-mz.124-15.T1.bin    50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4a.bin    33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin             5571584
12 : c2800nm-ipbasek9-mz.124-8.bin              15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin      33591768
14 : c2950-i6q412-mz.121-22.EA4.bin            3058048
15 : c2950-i6q412-mz.121-22.EA8.bin            3117390
16 : c2960-lanbase-mz.122-25.FX.bin             4414921
17 : c2960-lanbase-mz.122-25.SEE1.bin           4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin           4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin    8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin     10713279
21 : c800-universalk9-mz.SPA.152-4.M4a.bin      33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin      83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin     505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG        159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG        184530138
26 : ir800-universalk9-bundle.SPA.156-3.M.bin   160968869
27 : ir800-universalk9-mz.SPA.155-3.M           61750062
28 : ir800-universalk9-mz.SPA.156-3.M           63753767
29 : ir800_yocto-1.7.2.tar                      2877440
30 : ir800_yocto-1.7.2_python-2.7.3.tar        6912000
31 : pt1000-i-mz.122-28.bin                     5571584
32 : pt3000-i6q412-mz.121-22.EA4.bin           3117390
33 : sampleFile_FTP.txt                         26
ftp>
```

6.3 Step 3: Delete the file

I reconnected to the FTP server and removed the file:

Command used to delete the file:

```
delete sampleFile_FTP.txt
```

Then exited:

```
quit
```

```
C:\>ftp 209.165.200.226
Trying to connect...209.165.200.226
Connected to 209.165.200.226
220- Welcome to PT Ftp server
Username:student
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>delete sampleFile_FTP.txt

Deleting file sampleFile_FTP.txt from 209.165.200.226: ftp>
[Deleted file sampleFile_FTP.txt successfully ]
ftp>quit

221- Service closing control connection.
C:\>
```

7. Understanding FTP Concepts

FTP Login Process

FTP first establishes a control session over port 21. This channel handles commands such as listing files, renaming files, or initiating transfers.

Uploading (put)

The **put** command sends a file from the local PC to the remote server. FTP temporarily opens a **data channel on port 20** to transmit the file contents.

Downloading (get)

The **get** command retrieves a file from the server and downloads it into the PC. Again, a separate data channel is opened to ensure smooth transfer.

Renaming and Deleting

FTP supports file management commands:

- **rename oldname newname**
- **delete filename**

This gives the client full control over files stored on the server.

8. Conclusion

This lab provided a clear understanding of how FTP functions and how files are managed between a client and a server. I practiced authentic FTP commands, explored how files move across the network, and saw how FTP separates command communication from data transfer. These skills are essential for network operations, server administration, and managing configuration files in real environments.